Few-Shot Learning Approach to Brain Tumour Classification

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April 2025

Abstract

This project aims to use few-shot learning for brain tumour classification from MRI scans. The diagnosis of brain tumours can be difficult, so an AI model that can give a second opinion after a doctor would be very useful. Prototypical learning is the few-shot learning strategy implemented alongside transfer learning to classify an image given only a few labelled samples of each class. This is relevant as medical image datasets, such as the brain tumour dataset, are only small and not adequate for training a full network. This experiment was implemented in Google Colab using Python and used ResNet34 feature extraction. Gaussian noise and masks were applied to the dataset to implement causal learning and ensure that the model only learned representations from the causal features. The final model was a 3-way, 5-shot design, and many of the hyperparameters had been tuned through different optimisation methods. The model recorded an accuracy of 97.44% across the test images, which is comparable to many other designs created in similar research showing the applicability of few-shot learning for this task. The model could be further improved with a greater analysis of the dataset and improved fine-tuning methods, as well as a newer version of the ResNet pre-trained model such as ResNet50V2.

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1 Project Specification, Motivation and Aims

1.1 Introduction

Brain tumour classification is an important step in diagnosing a patient and continuing with the appropriate treatment. Brain tumours remain one of the deadliest forms of cancer with glioblastoma diagnosis proving to be fatal within 2 years for more than two-thirds of adults [1]. Some tumours can be treated with chemotherapy or radiotherapy but some like glioblastomas or other gliomas are resistant to both [6]. Therefore, the accurate classification of a brain tumour is vital for treatment.

This research plans to tackle the problem of brain tumour classification in deep learning through the use of few-shot transfer learning and prototypical networks. The project will create an accurate deep learning network to classify 3 different kinds of brain tumour provided by a dataset containing MRI scans taken at 3 different angles (axial, coronal and sagittal). Using the proposed prototypical network approach, the model aims to distinguish between these three classes using just the causal factors and not the background noise in the MRI scans. Few-shot learning is not a very well explored approach when it comes to brain tumour classification in deep learning. It has many advantages such as not needing many labelled samples to learn how to classify a tumour whilst also being capable of identifying new classes without needing to retrain the whole model. These advantages are useful in medical instances where labelled images are limited in quantity or hard to find. Therefore, few-shot learning is a suitable approach towards brain tumour classification and needs more research to be undertaken.

Transfer learning aims to improve the performance of a neural network on a target domain by transferring the knowledge in a different but related source domain [20]. In an ideal world, there would be a large amount of labelled data with a fair distribution of classes. However, this is very difficult to obtain in the real world as collecting all this data can be time consuming and expensive. Therefore, the solution is to utilise the knowledge from other domains. Semi-supervised and unsupervised learning can reduce the need for labels but sometimes it is hard to obtain a large quantity of unlabelled data such as medical imagery. These limited datasets make it hard to create a model which outputs accurate predictions. An incorrect classification is considered very bad in the medical world where diagnosis of the correct disease or illness is important in determining the next steps and suitable treatment.

As well as for limited datasets, transfer learning is used to enhance the performance of a model through previously gained knowledge, and to shorten training time. Some traditional methods for transfer learning include fine-tuning and frozen layers. Freezing layers involves using pre-trained layers from a different model without changing them. This is useful as you can use the knowledge from models trained on high quantities of data that are more effective at tasks such as extracting features from images. Fine-tuning consists of taking these layers and adjusting them slightly to fit better to the target domain. Sometimes the source domain used for pre-training is different but related to the chosen target domain so some adjusting of hyperparameters can help improve the performance and suitability to the chosen topic.

Few-shot learning (FSL) is a transfer learning strategy where a few labelled images of each class are used to define a class and then the new inputs are classified based on those few samples. The few-shot learning algorithm is based on the idea that machines should be able to copy humans and learn what an object is from only a few labelled examples. In few-shot learning, data is split into a support and a query set, and the support set is picked to be used during classification. The support set defines the classes and consists of a few samples per class; for example, a 5-shot classification would have 5 labelled images per class in the support set. This is where few-shot learning differs from zero-shot learning (ZSL) and one-shot learning (OSL). Sometimes one-shot learning can lead to poor generalisability as it is fully dependent on the labelled image chosen for that class. If that labelled image is poor quality or the predicted image is poor quality, this could lead to incorrect classification. Few-shot learning has some advantages over these other transfer learning strategies as it is less reliant on the single support image per class. It also results in higher classification accuracy than zero-shot learning due to there being more labelled support samples. These multiple labelled images for each class allow few-shot learning models to rapidly generalise new tasks with only a few supervised examples [17]. The number of classes used is written as N-way such that a 3-class problem would be 3-way.

1.2 Motivation

Magnetic Resonance Imaging (MRI) is the standard for the diagnosis and follow-up of primary brain tumours in adults [15]. Medical imaging does not always appear clear. Some machines may be better than others, and some may produce blurred images, especially if there is any movement by the patient while the images are being taken. Also, current techniques are already used to attempt to improve the image quality, so an artificial intelligent network which could accurately classify these brain tumours would be optimal.

Medical images are often scarce and difficult to find large datasets of. They are expensive to further increase the size of, due to the costs of resources to produce these images, such as MRI machines. It is also expensive to increase the size as you require a lot of patients with the illness and each patient also needs to consent towards their data being used for research. Traditional deep learning approaches may present many issues in creating an accurate model as a result of the limited quantity of medical images, especially relating to labelled samples of brain tumours. Traditional approaches are reliant on large datasets to train the model, but this is not applicable in medical scenarios. Therefore, there is a need for a suitable artificial intelligence (AI) model which can work and produce accurate results even when the dataset is small. This is why few-shot learning is suitable for this scenario as it can work effectively on small datasets.

Traditional transfer learning strategies are not as effective when it comes to identifying new unseen or barely seen classes. This is the advantage of a few-shot learning approach. There are more than 100 different types of brain tumour [3] and many brain tumour datasets only contain a select few brain tumour categories within them. As mentioned above, this is a result of medical images being difficult and expensive to obtain, especially for more rare diseases and diagnoses. In addition, many of these tumours may appear similar, so for a human to distinguish between them and accurately label them may be hard especially in the case when the MRI scan works poorly and the resulting image appears unfocused or contains noisy data where many of the tumour images look alike and have different levels of zoom or brightness as can be seen in Figure 1. Therefore, there is a need for a deep learning approach which can accurately generalise with only a few labelled samples. This is one of the advantages provided by few-shot learning as it only needs a few samples to understand what something is and classify an input query item to the newly defined class. Other traditional approaches and transfer learning strategies may require a large amount of data from the new tumour type to be able to classify into this new tumour category.

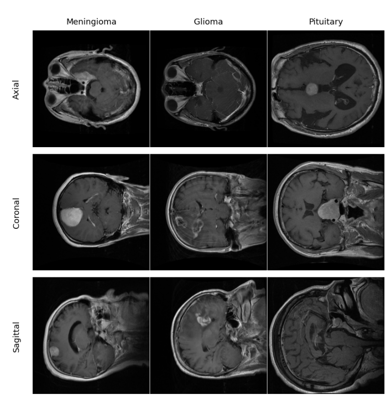

Figure 1: Grid showing the different types of tumour and planes taken from an MRI scan from the dataset provided by Jun Cheng on figshare [4]

Few-shot learning is an underexplored approach, especially for medical imaging. There is a gap in the research for few-shot learning approaches for neuroimaging and brain tumour diagnosis despite its many important features which can be utilised to solve many of these difficult issues. With many more types of tumour present in MRI datasets, there should be more research on developing transfer learning approaches that accommodate fast learning to new tumour types with only a few samples.

1.3 Aims & Objectives

This study aims to apply few-shot transfer learning to brain tumour classification from MRI scans and assess the effectiveness of using prototypical networks to implement this approach. Performance metrics such as accuracy, F-1 score, and confusion matrices will be calculated to analyse the performance of the model with test accuracy being the most important as it is being used to calculate the performance when altering the hyperparameters such that the best set of hyperparameters are assigned. This study will discuss the advantages and disadvantages involved in this method as well as any limitations found during research or carrying out the experiment as well as any future improvements applicable. This project aims to generalise well, despite the small dataset size and only using a few labelled examples. This is with the aim of working in clinical hospital scenarios where new tumours may be added and there are more than just the types of tumours present in the dataset. Some of these tumours may only have a handful of limited labelled images so an appropriate model for brain tumour classification would be one in which only few labelled samples are required to classify new input images to.

The project will be compared to other traditional approaches and the performances of each will be analysed. Different hyperparameters will be tested such as the pretrained model and number of shots to find the optimum hyperparameters for this task. The proposed model will then be compared with other created deep learning approaches to brain tumour classification designed by other researchers and see how their work compares as well as how their approach performed and if there were any differences in how the data was handled. The models will be compared in terms of their accuracy across the figshare dataset and how it performs with respect to its test accuracy and other metric scores.

This report will start with an analysis on the design, methodology and implementation of the proposed few-shot learning solution to brain tumour classification. This section will analyse different few-shot learning methods such as matching networks, Siamese networks and others and discuss what would work best in this scenario in the conclusion. A conclusion on the proposed few-shot learning approach will follow with reasoning behind why this approach was chosen and how it works. The research will also discuss the database chosen and the reasoning behind why the database is preferred to others, but it will also go over different preprocessing steps applied to the dataset to improve performance and ensure the model is only classifying based on the causal factors instead of the noncausal factors. The following section will discuss the implementation and methodology of the model and all the hyperparameters used as well as the environment used in these experiments. The different performance metrics will be discussed and given an explanation as to why these metrics were chosen to evaluate the model's performance. The experimental results section will discuss the results and graphs outputted by the experiments after running the tests on the proposed model. This section will visually show the performance metrics as well as evaluate what they suggest, and any potential improvements or useful information conveyed by them. The discussion section will mention some of the errors and misclassifications in the model and will aim to analyse why these images could have been misclassified or if there was any bias found when experimenting because of the imbalanced dataset. The project discussion section will also evaluate these experimental results deeper and compare these results with those of related works and compare the strengths and weaknesses of the proposed solution. This section will also address any challenges and limitations faced during development and any room for improvement which can be applied in future work. A conclusion will end this report detailing all the report and what the outcomes of the determined model are.

2 System Design

2.1 Few-Show learning Overview

Few-shot learning works in a similar way to one-shot learning, but more labelled images are used for each class instead of only one. Suppose that you gave a child multiple images of each animal from different angles or lighting, the child would be able to identify what the animal is if they were given another image of the animal shown. Providing multiple angles and lighting would make it easier for the child to picture what makes the animal that animal and therefore lead to better guesses. This is the same concept used in few-shot learning.

Sometimes one-shot learning can lead to poor generalisability as it is fully dependent on the labelled image chosen for that class. If that labelled image is of poor quality or the predicted image is poor quality, this could lead to incorrect classification. Few-shot learning attempts to decrease this dependency by supplying the model with multiple labelled images for each class as few-shot learning can rapidly generalise to new tasks with only a few supervised examples [17]. Suppose that you have a set of labelled images C for each class. You could compare the similarity between the input value X and every value ci where $ci \in C$. Then you could calculate the total similarity between for each class in the training dataset $f(x, ci)$. This calculation could then be used to classify X into one of the classes or potentially a new unseen class.

Few-shot learning is suitable for tasks with little labelled data, but can be used to improve the performance and generalisability of one-shot learning tasks. Few-shot learning is still dependent on the quality of the chosen labelled examples, but there is less of a reliance. However, this improved independence makes it not as well equipped for classifying to unseen classes.

This is useful for the brain tumour dataset due to its ability to still classify into unseen classes whilst still maintaining a good classification accuracy. Overfitting can occur due to the previously mentioned transfer learning strategies as the dataset uses MRI scans, which can look similar and could be matched based on the quality of the images. Also, the dataset uses MRI scans from multiple angles, so a few-shot learning would work well to use all angles to make a prediction in comparison to one-shot learning.

2.2 Few-Shot Learning Approaches

There are many different strategies for few-shot learning such as matching networks, prototypical networks, Siamese networks, relation networks, and model-agnostic meta-learning (MAML).

Matching networks are one of the most common few-shot learning approaches [9]. Matching networks work by comparing the input with each vector of each class. The distance score for each class is calculated using a distance calculation method such as Euclidean distance, and then the mean is found for each class. The closest class average to the input is therefore the given class for the input value. The paper on few-shot learning through prototypical networks [9] states that simple baselines can outperform meta-learning methods by using embeddings and standard cross-entropy loss function.

Prototypical networks work by creating a prototype of each class and then computing the distances between the input and the prototypical representation of each class [13]. These networks work by finding an embedding around which points of a class cluster. This is done by computing the input into a non-linear matching then taking the prototype of the class to be the mean of the data in the embedding space. Classification is done by finding the nearest prototype. Prototypical networks are a more efficient and simpler approach than meta-learning [13]. Due to only calculating the prototype once and then comparing each input with only the prototype, this means it is less computationally expensive than matching networks [16]. In one-shot networks, both prototypical and matching networks are the same. A paper on prototypical networks [13] stated that using the squared Euclidean distance can greatly improve the performance over other distance calculations for both prototypical and matching networks. The paper on prototypical networks for few-shot learning [13] created a good figure to represent how prototypical networks work. Figure 2 shows how this can work visually by creating a prototype of the 3 classes given

5 support images for each class. This is a 3-way, 5-shot prototypical network. The distance is then calculated from the input image to each prototype and assigned the class of the closest prototype.

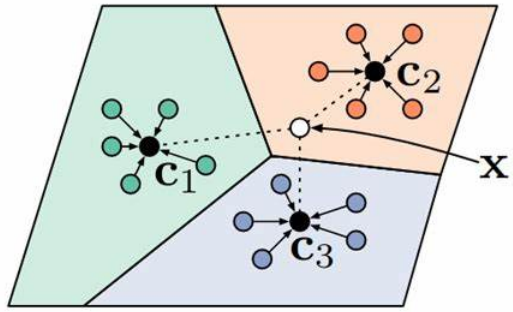

Figure 2: Taken from a paper by J, Snell et al. [13] visually displays a prototypical network with the c value black dots representing the prototypes formed from the inputting circle support set embeddings. The circle labelled X represents an input of the query set which is being assigned a class based on the closest prototype. This is shown through colours to show what the bounding ranges for each prototype could look like.

Siamese networks can be beneficial to use for comparison if the two objects have different dimensionalities or types [5]. Siamese networks consist of two artificial neural networks, where they both learn the hidden representation of the input vector. They are both feedforward perceptrons and use backpropagation in training. The output for the similarity is then calculated by both artificial networks, often using a cosine distance function. This is the meaning behind the name Siamese, as the two artificial networks need to be identical and share weights. For image data, CNNs are used as the Siamese networks. At the end a binary output will be returned to say if the input is the same class or not. Therefore, a popular loss function for the Siamese networks is binary cross-entropy. The intention behind Siamese networks is to keep the same classes as close as possible to each other and different classes as far away from each other within the embedding space [16]. The inputs are passed through two identical networks before passing through a contrastive loss function to form the feature map of the metric space. In the study [16] on plant leaf classification, the test images are passed into this feature space before calculating the k-nearest neighbours through the Euclidean distance and then categorising by each class's score. Due to their double network format, Siamese networks are popular to use when the task is binary, such as true or false, or for one-shot learning tasks.

Relation networks work by reasoning with the relationship of objects between images. These are often associated with spatial relations, such as where the object is present in the image. Relation networks are useful for object detection because of their ability to recognise relationships. They work by not just spatially in the image but also relative to other objects in the image. For example, it could find a chimney by knowing that it is always above the house. Relation models may sometimes be difficult to implement if the objects are not always in the same place relative to each other [9].

Model-agnostic meta-learning (MAML) can be difficult to train for few-shot learning scenarios [19]. However, MAML is one of the best meta-learning approaches to few-shot learning [2]. The idea of MAML is to achieve a high generalisation through applying a small number of gradient steps. However, MAML suffers from many disadvantages, such as the need for hyperparameter tuning, costly overhead, and reduced framework flexibility [2].

2.3 Model Selection

The chosen few-shot learning method is prototypical networks. Prototypical networks work by creating a prototype of each class using the support set, where the prototype is the average of all the embeddings. Then, when classifying an image in the query to the support, the image is assigned the category of the nearest prototype. This works quicker than matching networks in which each query image is mapped to every value in the support set of a class, and the distance is calculated to which the class with the smallest average distance is assigned. Prototypes only calculate the distance between the prototype and the input by calculating the prototypes at the start, removing the need to calculate the prototypes again in the future and only needing to calculate the distance the number of classes times instead of the number of classes multiplied by the number of shots. Matching networks can be quite expensive when using more shots.

Furthermore, a paper focusing on few-shot learning approaches [13] compared prototypical, matching and relation networks in both one-shot and 5-shot scenarios. Relation networks performed best in one-shot learning, but prototypical networks performed better in 5-shot tests. Prototypical networks are also the cheapest in terms of computational cost due to there being only one comparison between the input and each class. Prototypical networks also learn class representations directly, unlike Siamese networks, which need explicit pairwise training. Relation networks require more data than the other few-shot learning approaches and may not be as effective due to the random positioning of the brain tumours throughout the dataset, as each tumour is not fixed to a specific location on the brain.

2.4 Dataset Selection and Preprocessing

This experiment will be using the brain tumour dataset posted on figshare by Jun Cheng [4]. This dataset consists of 3,064 MRI images from 233 patients with three different kinds of brain tumour. Therefore, there are three classes which the model will attempt to classify, making this a 3-way few-shot learning model. These three tumour types are pituitary, glioma, and meningioma. There is an imbalance in the dataset and the representation of each tumour category within it such that there are 708 meningioma, 1,426 glioma and 930 pituitary images. These images are clinically valid as they have been taken from Nanfang Hospital in Guangzhou, China and the General Hospital at Tianjin Medical University. This is a useful dataset for multiple reasons with the first being that it provides MRI scans at different angles. This is representative of real-world scenarios as the tumour may not be visible from some angles so it is important to be categorised from different scan angles. These different angles and tumour types are represented in Figure 1 where the different places are axial (from above), coronal (from the front), and sagittal (from the side). Secondly, in terms of medical imaging, this is a larger dataset than others, which are often restricted in size. The large size allows for enough images for training but then also enough for validation and testing. The final important feature of this dataset is its masks. Each data entry is matched with a label and a mask. These masks map where the tumour is on the image by creating a separate image of black and white pixels where the white pixels indicate where the tumour is located. This can be utilised for adding noise to the model such that the model does not categorise an image based on some irrelevant feature in the background. Instead, the noise will force the model to not learn these background objects as a part of the human's head but instead looking and extracting the tumour directly.

For this dataset, images are resized to fit within RAM compliance. Different image sizes were tested and 256x256 was the best performing image size as it stood in a middle ground between the original image dimensions of 512x512 and the ImageNet dataset size of 224x224 that ResNet34 was trained on. Many data augmentations and random transformations were used to improve generalisability and decrease overfitting. Random rotations, flips, and affine were applied so the model does not generalise by the direction or orientation of the original images. As well as that, some random colour jitter was implemented with small values of contrast and brightness as to replicate the MRI screening and not generalise based on the quality of the image taken. This also makes it better for translating its learning over to other datasets in which the images may not be as clear or focused.

The training process also utilises noise to focus the model onto the correct part of the image and aid in extracting the correct information that is relevant in determining the tumour type. This uses the previously mentioned masks in which Gaussian noise is added to the background of the image where the tumour is not represented on the mask. This is shown in Figure 3 where the original, mask and noisy images are displayed to show how the Gaussian noise affects the model. Through using masks and noise, this algorithm implements causal learning as the model focuses on what causes the labelling of the tumours rather than any background non-causal features such as other features of the brain or eyes.

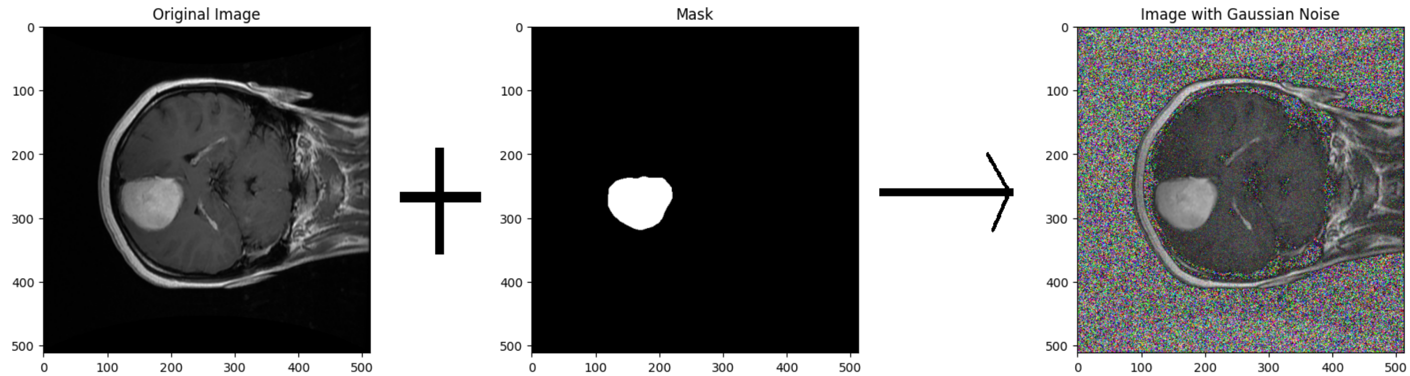

Figure 3: Image showing how Gaussian noise is added to the image through showing the original, unfiltered image, the black and white mask displaying the region of the image where the tumour is and the final image of the noisy image after Gaussian noise has been applied.

For training, validation, and testing; different subdatasets were used such that no overfitting or biased results could occur as a result of reusing the same data. To do so, there was a 70:20:10 split between training, validation, and testing, respectively. In addition, the dataset kept similar data images congregated with each other such that images of a specific class and on a specific plane were all adjacent to each other in the dataset. Therefore, a shuffle was applied so that not all of the data was put into one sub-dataset or the other. To allow for reproducibility while testing, a seed of 42 was applied so that it shuffles into the same order each time. This allows testing to work effectively and compare different hyperparameters without worrying that the changes in accuracy could have been caused by different images being present in the training/validation/testing datasets.

3 Methodology and Implementation

3.1 Development Environment

This proposed solution was created and run in Python 3.10 on Google Colab. Google Colab was used due to its simplicity and ability to create multiple versions of the project for hyperparameter testing at the same time by opening multiple windows with Google Colab on each one. However, at the same time, Google Colab has slightly hindered the development of the project. The RAM restrictions and size on Google Colab are only 12.7GB with a limited use of the GPU RAM. Therefore, only some tests could be run using the T4 GPU and most had to be run without it. This played a limiting factor in certain features and hyperparameters that could be implemented in the model. For example, Google Colab would not allow for higher pre-trained network versions such as ResNet50 due to crashing the runtime once the system ran out of available RAM space. Similarly with image sizes, they would not allow for the original image dimensions of 512x512 so the images had to be reduced and could not be tested at full size.

Pytorch and torchvision libraries were used to develop the prototypical network and provide data augmentations to the dataset. Pytorch was chosen because of its popularity in creating neural networks, but also because it allows for easy creation of the prototypical network by inheriting the neural network module. However, Pytorch was also used for creating the dataset from the original ".mat" files. Tools such as NumPy were used to handle the data, and results were visually plotted with the help of Scikit-learn and matplotlib such as the confusion matrices and classification tables which were implemented through Scikit-learn. Matplotlib was used to plot the line graphs and visualise aspects such as T-SNE and the validation/training loss and accuracy curves. T-SNE was also implemented through Scikit-learn due to its built-in, easy to implement library functions. As mentioned, random was used to shuffle the dataset so that similar data was not clustered together with each other, but rather spread out throughout the dataset. The random seed used to implement this and allow for reproducibility was 42.

3.2 Methodology Overview

This is a multi-class classification problem where there are three classes in the chosen dataset. For this task, the model needs to learn the differences and defining features of each tumour type to distinguish between them at different projections. Therefore, few-shot learning is an appropriate approach to create a potential solution. Few-shot learning works well with limited data which is the case in the instance of medical imaging with as stated only 3,064 images of each class. As well as that, few-shot learning is well adapted at learning new classes such that if any new classes were added, it would not require re-training the whole network again, as is the case with traditional approaches. It can also learn these new models with minimal data, so the new classes do not need many labelled sample images to form an effective support set. The support sets are the sets of samples which the model makes the predictions from. The query sets are the input images that the model classifies to one of the tumour classes. This is done through calculating the distance of the input query sample to the support set of a specific class. These distances are reversed to give them a score so that the highest score represents the most suitable class and hence the class it is predicted to belong to. If all these scores are low, it may assume that the input query sample belongs to a new unseen class. This shows its importance and utilisation in working with minimal datasets, as only a small number of support set samples is required to make an effective classification set.

The chosen few-shot learning approach is a prototypical network. This few-shot learning approach was preferred to other methods as it only relies on one distance calculation between each prototype rather than every sample of the support set like matching networks. In addition, relation networks are dependent on the location of features in an image. This makes it harder to train in instances like brain tumours where tumours of the same type can appear in different locations of the brain. Prototypical networks work by creating a prototype of a support set by calculating an average for each class and using that as the prototype. The distances are then only calculated between the input query image and the three prototypes of the support set. The prototypical network is created by inheriting the neural network module class from Pytorch. Then specific attributes and methods such as the backbone, loss function, distance calculation, and forwarding functions need to be determined and produced. The produced prototypical network is expected to classify for 3 different classes of brain tumour where each support set will contain 5 images of each class (it is a 5-shot network) and 15 query images of each class. Then the prototypical network will form feature embeddings from the support set such that it can create a prototype by calculating the mean of the embeddings. These prototype embeddings are then used to form predictions on the query images.

An appropriate distance function is vital in determining the closest prototype to the input query image. A paper on prototypical networks [13] stated that using squared Euclidean distance can greatly improve the performance over other distance calculations for both prototypical and matching networks. Therefore, the squared Euclidean distance seems like a suitable distance function.

The loss function being used is focal loss as this is suitable for class imbalances which as stated this dataset contains. This additionally resulted in a better performance outcome compared to other loss functions like cross-entropy when tested.

The prototypical network will be using the ResNet34 backbone for feature extraction. ResNet is well suited for medical imaging [18] such as the MRI scans being used. However, due to limited computational resources, this model will only be using ResNet34 rather than ResNet50 or ResNet50V2. As the model will be using a less recent version of the pretrained model, the overall performance of the model may not be as good. This would be due to the older versions not being as effective at feature extraction as the newer versions and any slight error when extracting the brain tumour from the MRI could lead to a misprediction if the wrong features were shown. The prototypical network will use some fine-tuning to improve the feature extraction and target it more towards the target domain. To do so, two additional layers were added as well as a dropout layer to avoid any overfitting caused.

The Figure 4 visually shows how the initial creation of the prototypes is implemented and what layers the support set are passed through to create the prototypes. The support set images are passed through these layers, and a mean is taken of the support set image embeddings of the same class such that the prototype is formed.

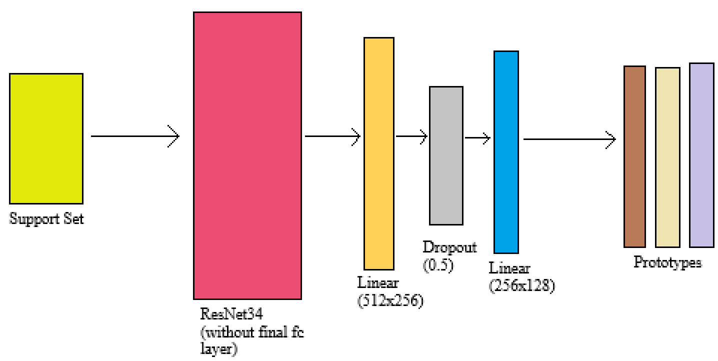

Figure 4: The design of proposed prototypical network for forming the prototypes. Shows the support set images being passed through the transfer learning backbone with the final fully classified layer removed before being passed through some fine-tuning layers and then averaged to create the prototypes.

3.3 Training

Due to this being a few-shot learning approach, episodic training is the best method for training the network. Episodic training is a popular approach to training few-shot learning networks such as the chosen prototypical network. Episodic training works through organising training into a set of learning problems (which are called episodes) where each episode is split into a small training and validation set [9]. For each episode, 5 support data items and 15 query data items were used from each class meaning there were 60 images total used per episode. This allows for a true prototype to be learned along with the embeddings, as the whole training set will have an attempt at being both the support and the query sets. This avoids relying on the selected support set which may not accurately represent the whole dataset. This is shown by the fact each episode used 60 images, and the training ran 100 episodes per epoch. To evaluate which model was preferred, accuracy was used with a combination of confusion matrix to analyse the performance and check for any bias or overfitting caused by changes to the hyperparameters.

Many hyperparameters were changed to find the highest performing network. These hyperparameters are features such as the number of epochs, number of episodes per epoch, optimiser, learning rate, number of shots, and the gamma value of the focal loss function. Some of these hyperparameters were optimised using grid search such as testing the learning rate. Meanwhile the majority were tested through trial-and-error with the testing accuracy produced being the independent factor on which hyperparameter values were employed. Other hyperparameters were chosen as a result to solve issues with the model such as overfitting. The focal loss and dropout rate is an example of this as the original cross-entropy loss caused high training accuracy but significantly lower validation accuracy. A combination of this and tuning other hyperparameters helped aid the goal of reducing overfitting.

Some optimisation methods were implemented to find the best value for the hyperparameter. Nelder-Mead optimisation was used to find the best number of shots available for the network in which it determined 5 was the optimum. The Nelder-Mead optimisation algorithm is designed to solve the problem of minimising non-linear functions. This optimisation algorithm works by evaluating the function at the vertices of a simplex and then shrinking the simplex based on the results of the function [12]. This is repeated until an acceptable simplex dimension is formed at which the best hyperparameter values have been found.

To implement the training, 50 epochs were used with early stopping with a patience of 5 added to prevent any overfitting and produce the best model outcome from training. This used the validation loss as a basis, meaning that if the validation loss did not increase within 5 more epoch iterations, the training would finish and the model would revert to the best performing model before the 5 patience epochs which performed slightly worse. Each epoch ran for 100 episodes.

3.4 Success Criteria and Performance Metrics

For testing and evaluating the performance of the model, episodic testing was used, similar to training. However, the number of episodes was reduced to 20 as the testing dataset size is much smaller. The tests were similarly implemented using a 3-way 5-shot learning approach to simulate the training and represent the dataset fully. To create the testing set, a split was implemented to separate the data into training, validation and testing. The test set was given 10% of the overall dataset or 305 images total. A fixed testing dataset was used rather than k-fold cross-validation as the model took a little while to train in the Google Colab environment, especially as it would run 100 episodes per epoch and would struggle to handle this when only run on the CPU.

Throughout training and testing, many metrics were used to analyse the performance of the model. During training, both accuracy and loss for a training and a validation set are taken into consideration and plotted on a graph to visualise and allow easier checking for any overfitting or errors in running. These also show if the hyperparameters are working well during training instead of having to wait until training is over to see if they worked. Accuracy is calculated as the number of true predictions / total number of predictions. This is important as it is simple to calculate whilst showing the performance of the model in a numerical value such that it can be compared to previous versions of the model with a different set of hyperparameters and can be evaluated and continued appropriately.

The test set makes use of F-1 score to analyse the performance of the model and metrics such as precision and recall. Precision is the number of true positives / (true positives + false negatives) while recall is the number of true positives / (true positives + false positives). A low precision implies that it is being biassed towards that class as it is predicting positively for it more than is true. Meanwhile a low recall implies the model is not predicting the test set suitable and it tends to classify the input images to that of different classes than the true class. The F-1 score is the average of both precision and recall as both values are important to be high, especially when misclassification can be so detrimental.

The confusion matrix can also help visualise this and show if there are any confusions between the class predictions and if the image embeddings need to be separated more. This is because the confusion matrix can show where the model may confuse tumour types by showing numbers of predictions for each class compared to their true value. This can also be represented with colour to demonstrate if there are any areas of interest that need to be sorted.

High accuracy is important in the real world as incorrect predictions can be harmful with different diagnoses being administered depending on what kind of brain tumour the patient has. Therefore, it is vital to reduce the number of incorrect predictions. Whether they are false negative or false positive, they are equally harmful. However, if no tumour MRI scans were to be added to the model, it would be important not to falsely predict a patient who has a brain tumour as a patient who does not as it is important to catch a brain tumour early before they further infect or spread across the brain tissues.

The model should perform with a high accuracy above 90% as shown by previous models such as [10, 14, 11, 7] performing with accuracies 91.43%, 99.68%, 99.6%, and 98% respectively. However, some of these models were tested on a higher version of the pre-trained network so the accuracy may not reach as high as them. To test the base accuracy, the pre-trained network ResNet18 was tested with no fine-tuning or few-shot learning implemented but rather just changes to the classifier layer to predict one of the three tumours instead. This resulted in an accuracy of only 73.02% which already proved to be worse than ResNet50 which tested at 74.60% with no fine-tuning applied. Therefore, the aim of this research experiment is to create a few-shot learning approach to 3-way brain tumour classification which has an accuracy ¿90% as it should much outperform the unchanged pretrained network which has not seen or been trained at all on brain tumour MRI scans.

4 Results, Evaluation, and Testing

4.1 Experimental Results

Many different experiments were run to test different hyperparameter combinations. Different optimisation methods were used to find the best combination of hyperparameters such as trial-and-error, grid search, and Nelder-Mead optimisation. Some experiments that were conducted were running different pretrained networks, image sizes, different loss functions, learning rates, percentage of training images with masks applied, and number of shots.

The number of shots is an important feature to test as this is representative of what the prototypes that decide classes will be. If there are too few shots, the support set will struggle to create a good prototype that is representative of all data. But if too many are selected, this can cost a lot of data which may not be necessary or may take too much data such that the query set testing the support may not be able to be as big. Therefore, the Nelder-Mead optimisation was run to find the best number of shots for the model which finds a middle ground between too few shots and too many shots. The original, starting value input was 5 shots and this ended up being the outcome of the algorithm as it returned 5 shots as the best number.

The pretrained network is also important to select the best version available to improve the performance of the model. As mentioned, the limited RAM provided by Google Colab restricted the ability to use higher versions so other versions had to be tested and ResNet50V2 could not be assumed to be the best transfer learning model for this dataset given the research done by Md. Alamin Talukder, et al. [14]. Therefore, different pretrained networks had to be tested such as ResNet18, ResNet34, and ResNet50. ResNet34 was the better performing of the two older versions whilst also still running under the RAM restrictions and small size unlike that of ResNet50.

Different image sizes had to be tested to find a good middle ground between the 224x224 images that the original ResNet model was trained on and the 512x512 size of the MRI images from the dataset being used. A range of image sizes was tested from 212x212 to the original 512x512. These tests created a sort of bell curve around 256x256 in terms of accuracy, with 512x512 crashing the systems operation due to using up all the available memory. The image sizes tested were: 212, 224, 256, 312, 412, 456, and 512 and as mentioned only the latter had failed to execute. The remaining image sizes returned accuracies: 85.00%, 87.50%, 90.83%, 87.50%, 85.00% and 69.17% respectively. This accuracy creates a bell curve with 256x256 as the middle and highest accuracy.

Different loss functions were tested such as cross-entropy loss and focal loss. These were tested through trial-and-error with focal loss proving to be slightly better in performance for the architecture. The cross-entropy loss function tended to overfit and cause a difference in the training-validation accuracy graphs. Therefore, the loss function was changed to focal loss, and this reduced the gap between the two curves.

The percentage of images with masks applied was tested using trial-and-error where four different percentages were tested. These percentages tested were 0%, 50%, 75% and 100%. Different proportions of masks were applied so that the model did not become reliant on a mask whilst also using the mask to identify the areas of importance. This way when the real test images were applied, the model would not become confused and identify tumours in locations where there are not any purely because they look different in testing. The resulting accuracies from these tests were: 94.78%, 96.44%, 96.78% and 96.22% respectively.

Learning rate was another hyperparameter that had to be tuned. The learning rate is important for making sure it reaches its true highest accuracy without hitting a certain accuracy then dropping straight away as is the case with high learning rates, or taking too long as is the case with small learning rates. The early stopping was altered so that the patience had a value of 10 for these tests. This ensured that if the model was still learning after 5, it would not be stopped due to a high validation loss in one of the earlier few epochs. The three learning rates being tested were 1e-4, 5e-5, and 1e-5 and the accuracy was measured for each instance. The accuracies achieved were 96.78%, 97.56%, and 96.11% respectively. This showed that the best learning rate for this project was 5e-5 (or 0.0005).

The overall performance of the final model design was evaluated using classification tables and confusion matrices from Scikit-learn but also the training and validation loss/accuracy curves were plotted to show the performance of the model. The test set consisted of 305 images with 70 coming from meningioma, 142 glioma images, and 93 pituitary. These images were tested using episodic testing such that each episode contained 20 images of each class. Therefore, different sets could be tested as the support set and the query set in case there was any bias in what support set images were chosen in this instance. The small number of images for the support set make it more practical for the real world where there are not that many labelled images, if any, of every type of brain tumour. In the end, the accuracy returned a value of 97.44%. The statistical significance was calculated for these experiments to prove the performance of the mode. To do this, episodic testing was used with 20 different combinations of 5 support and 15 query images. The resulting accuracies provided an average of 97.44% with a 1.69% standard deviation and a P-value of 9.31e-32.

The classification table consisted of the F-1 score, precision and recall values for each class. This is important as it allows visualisation of how each individual class is performing and shows if there is any bias or neglect in certain classes. This also evaluates the overall performance of the model with numerical values such that it can be compared to other designs either to test previous proposed models by this research or to compare in depth with other studies in related works that similarly worked with a deep learning approach for brain tumour classification. This classification table is depicted in Table 1 and shows the performance metrics output by the model over the test set.

Class	Precision	Recall	F1-Score	Support
0	0.9695	0.9533	0.9613	300
1	0.9554	1.0000	0.9772	300
2	1.0000	0.9700	0.9848	300
Accuracy		0.9744		900
Macro Avg	0.9750	0.9744	0.9744	900
Weighted Avg	0.9750	0.9744	0.9744	900

Table 1: Classification table showing the performance metrics per class and average. Classification table also displays the recall, precision and f1-score performance metrics for the classes 0, 1, and 2 or meningioma, glioma, and pituitary respectively.

The confusion matrix is another way to visually see how each class is performing and how the model performs. The confusion matrix goes into more depth on what the predictions being made by the model are as it shows directly what is being predicted and what the true value is (what the model should have predicted). This can show if the model is confusing any of the classes which could demonstrate either mislabelling in the dataset or confusion in the model when differentiating between the two classes. The output confusion matrix from the final version of the model is shown in Figure 5.

Finally, the two loss and training curves have been plotted and shown for the final version of the

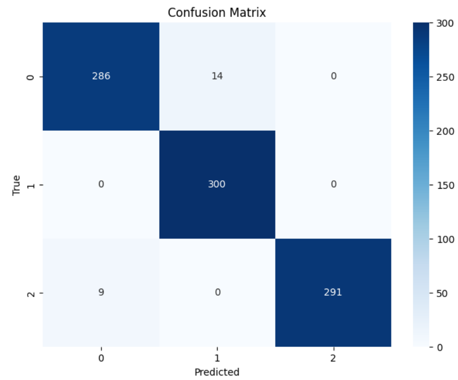

Figure 5: Confusion matrix showing the results and predictions made by the final version of the proposed solution. This shows a slight imbalance in the dataset and the model being more favourable towards glioma than the other classes.

proposed model. These curves display the training and validation values for both graphs and can display if there are any issues with the model training such as overfitting. These graphs also demonstrate how effectively the graph is learning as shown by the curve and how the gradient is. The two curves produced in this project are shown in Figure 6.

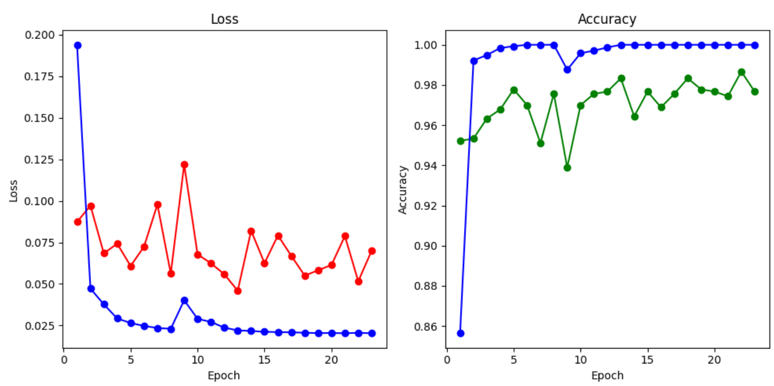

Figure 6: Loss curve (left) and training curve (right) showing the training and validation results from each epoch during training. Blue represents the training values; red is the validation loss and green is the validation accuracy. This shows successful learning from the model with only a small amount of overfitting present (as shown by the distance between the two accuracy curves.

4.2 Model Evaluation

These results outline successes or failures within the model. The preferred number of shots being 5 is surprising as it is only just about enough to cover all projections of the image. Therefore, in some of the support sets, some of the images may not have been present yet the model still predicts well. This suggests that the model is adapting well to the actual casual feature in the tumour rather than any noise or the shape and size of the tumour. This is useful as it shows how the embeddings are separated by their tumour type and the same type of tumour embeddings are close to one another suggesting that the embeddings have been created effectively. The newest version of ResNet is the most accurate as expected, due to it being a newer version, and the results outperformed the previous version. The favoured image size being 256x256 is reasonable as it is close to the source domain that the pretrained network was trained on whilst also not losing too much detail from the original MRI scan images, such as the defining features.

This follows other research into this topic such as J. Paul, et al.[10] who also concluded that the optimum image size for transfer learning with the figshare dataset is 256x256. The change in accuracy between the cross-entropy loss function and focal loss function suggests there was some overfitting beforehand as a result of the imbalanced dataset. The focal loss minimised the effect of the imbalance which therefore increased the testing performance of the model. The preferred learning rate of 5e-5 is suitable as the learning rate is not too high that it learns too quickly and reaches its ceiling too early. But also, the learning rate is not too low that it takes too many epochs to reach a good value of accuracy as was the case with 1e-5 which ended up running out of patience in early stopping.

The overall performance of the model producing a 97.44% accuracy implies the model is effective and works well at brain tumour classification from MRI scans. Using a significance level of 0.05, the very low calculated p-value suggests that these predictions were unlikely to have been randomly produced but rather the AI model recognised patterns and used these to distinguish between classes. Therefore, the result is statistically significant. The small recorded standard deviation across the 20 episodes further shows that the model isn't getting lucky in its predictions but rather it is consistently getting high accuracies across its tests. This standard deviation helps prove the performance of the model on the test set. The high accuracy and slight imbalance in the dataset could have been covered if there had been any bias in the dataset towards specific labels. Therefore, classification tables and confusion matrices were created to display the performance of the model for every class in the dataset. This high accuracy shows that it performs well and can accurately predict which class a tumour is in given a sample image of it from any plane. The accuracy passes the original success criteria of being greater than 90% and is much better performing than the 73.02% accuracy of the original model in which no fine-tuning nor few-shot learning was used.

The confusion matrix effectively shows how the model performed across each class of tumour. The recall and precision for all classes was above 95% showing there is little to no bias present in the dataset. However, there was still some bias. The confusion matrix shows how many predictions were made for each class with meningioma being the most predicted class at 314 of the 900 predictions compared to meningioma 295 and pituitary 291. These distributions are representative of the original dataset imbalances suggesting that there is a slight favourable output towards glioma if the model cannot decide if it is one of the other classes. The lowest f1-score was for meningioma showing that the model has not fully learnt how to differentiate meningioma from other classes yet. This could be due to limited amounts of data. If the model was provided with more, the model could learn the differences better and perform more like that of glioma. The best diagnosis tumour was glioma which had an f-1 score of 97.72%. This could have been influenced by the fact there was more data present for the glioma class than any other class. Therefore, the model would have seen more samples during training and is better adapted at predicting the glioma type. This would also explain why meningioma is the poorest performing of the classes with a 96.13% f-1 score. The glioma tumour class has the lowest precision out of all of the classes. This could suggest there is a slight bias towards the glioma dataset as it predicts more false negatives than any other class. Both other classes have a higher precision than they do recall which suggests there are more false negatives, and it is predicting a different class for when it is their class. However, the differences in the precision and the recall are so minimal that there is little to no effect of the bias.

The line graph representations of training and validation loss/accuracy show there is minimal overfitting in this model. The gradual increase in accuracy and decrease in loss while both starting off first with a steep gradient show that the model is learning well to the data and reaching a high performance. The bumps in the curves of the validation set imply that those specific sets contain difficult to classify images that are harder than those in the peaks of the validation curve or minimums of the validation loss curve.

4.3 Error Analysis

The model did not perform perfectly; there are some mispredictions in the test set. The correct diagnosis can be vital in medicine, so these incorrect predictions need to be reduced. One approach to aim to solve these solutions is to analyse what went wrong with the predictions and if there were any difficulties in the image. The incorrect prediction images were displayed such that this analysis could take place. Figure 7 shows some of the incorrect predictions in the test dataset. These images show that the model struggles when the tumour is not fully clear. These tumours appear faded and blend into the natural colour of the MRI scan and the background. For example, the third image in this figure is difficult to view as it blends in with the eye sockets. This suggests that more data samples containing these tougher images and more exposure during training is required to increase the performance of the tests. These tumours are a lot harder to notice than some of the other images in the dataset. However, the deep learning approach for brain tumour classification has the intention to aid and assist a doctor in classifying a brain tumour from an MRI scan. Therefore, it should perform well on the tough images as these are the kinds of images doctors are more likely to struggle with and need a second opinion for.

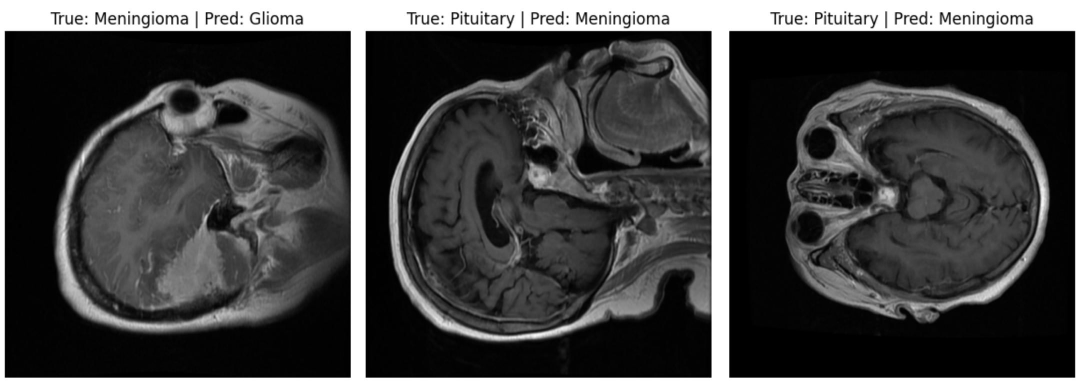

Figure 7: Three examples of failed predictions from the testing dataset when evaluating model performance. These images demonstrate that the model struggled when the tumour was small or hard to locate or that the model may have assumed some of the background noise was a tumour.

For these incorrect predictions, the confusion matrix and classification table show a pattern of favouring for predicting the glioma and meningioma class with 303 predictions for the meningioma class and 306 for the glioma class compared to only 291 predictions for pituitary. This suggests that a wider variety of images need to be shown for pituitary as it could be that the range of the prototype is only small. Adding more images of the dataset would improve this by widening the range of images seen and difficult images to predict will be labelled and the model should learn from them.

5 Project Discussion and Reflection

5.1 Discussion of Findings

This few-shot learning approach to brain tumour classification worked effectively in comparison to other traditional deep learning methods. But there are some other approaches attempted by other researchers to solve this classification problem.

J. Paul, et al.[10] similarly used the figshare dataset to train and analyse their model. They pre-processed their data slightly differently however and decided to only use data from one angle to make it simpler for their model. They used 989 of the axial images (images taken from above) instead of all 3064. In these tests, they used two different types of neural networks to analyse their performances and choose the best one. These two types were traditional fully connected networks and convolutional neural networks (CNN). They demonstrated that their chosen model could prove to be 91.43% accurate in their 5-fold cross validation tests for the CNN model. They also resized their images and found that 256x256 was the best performing for them. As shown by the experimental results, the few-shot learning network developed from this research proves to perform as accurately as this model despite utilising all images on both coronal and sagittal planes and not just the axial plane.

A more recent research experiment carried out by Md. Alamin Talukder, et al. [14] proved to be much more accurate. In their research, they also used transfer learning to improve the performance of the model similar to this research's proposed model. However, they utilised reconstruction and fine-tuning to better the performance of their model. During their tests, they tested different pretrained base models to find which worked best for them and they found that ResNet50V2 was the best performing of the 4 pretrained networks they tried (with IneptionResNetV2, Xception, and DenseNet201 being the other three networks). Unlike the previous related work, the researchers in this project used all images in training and testing so their data would have been identical to the data used in this investigation. This project proved to be much more accurate and better performing compared to the few-shot learning model created in this research. This suggests that the model proposed in this research should have more fine-tuning and reconstruction utilised before passing the backbone into the prototypical network. Also, the computational capabilities limited the project with Google Colab not allowing the use of ResNet50V2 but only ResNet18 due to the RAM limitations.

To compare the model proposed by this experiment, many other related works were analysed [10, 14, 11, 7] and studied with respect to the proposed solution as the summary shows in Table 2. Therefore, the proposed few-shot learning approach from this research could be useful for real-world brain tumour diagnosis due to high accuracy. The few-shot learning approach allows for new tumours to be added and included in the model and easily learn to fit to the new data. This is suitable for the real world as there are more than just 3 different types of brain tumour so a proposed AI model solution would need to be able to fit to all the real-world brain tumours. However, the model would need to be further improved if it were to be used in hospitals as false predictions could be very harmful so it would only be useful if it could be better at diagnosing than a real doctor. On the other hand, it could be a useful base step to give a second opinion on what type of tumour is present in the MRI scan.

5.2 Challenges & Limitations

There were many challenges and limitations encountered in the development and creation of this proposed solution and experiment. Firstly, the dataset is slightly imbalanced which could have resulted in overfitting and the model prioritising the highest popularity class and neglecting the other classes when it made its predictions. However, this was not the case as shown by the confusion matrix and performance metrics which represent a similar rate of predictions for each class. This imbalance might have been nullified because of both the data augmentations and transformations re-populating the data and reducing the bias towards one specific dataset as well as using focal loss which is designed with the intention of reducing the effect of an imbalanced database. This could also be as a result of the choice of few-shot learning architecture. Few-shot learning means that only a few samples are needed per class to create a prototype therefore despite the minimal data for some classes, there is still sufficient data to create a prototype and learn from it. As well as data imbalance, there were a few other challenges with the dataset, another being caused by the noise in the dataset. Some images appeared clear whilst others were distorted by the light from the MRI scan, or some were out of focus if the patient was not fully still when the image was being taken. This created a challenge as choosing the appropriate support set is vital for identifying the tumour type and if all the support sets for a specific class are clear or all the support sets are blurry, this can result in incorrect predictions as the model lacks detail or knowledge of blurry images. Therefore, as a solution, image augmentation was applied, and colour jitter was applied with a contrast of 0.2 and a brightness of 0.2 to make it similar. Finally, as mentioned earlier, this project was limited by the size of the dataset. Transfer learning was an approach to reduce the impact of a small dataset, but a larger dataset would be preferred.

There were also some issues with the few-shot learning model in overfitting. When the fine-tuning was applied after feeding the input through the ResNet backbone, the model was prone to overfitting and the

Researcher	Model Architecture	Accuracy	Image Projections	Pretrained Network	Training : Validation : Testing Split
J. Paul, et al. [10]	CNN	91.43%	Axial Only	None	78:11:11
Md. Alamin Talukder, et al. [14]	Transfer learning architecture with reconstruction and fine-tuning	99.68%	Axial, Coronal, and Sagittal	ResNet50V2	80:10:10
T. Sadad, et al. [11]	U-Net Architecture	99.6%	Axial, Coronal, and Sagittal	NASNet	80:00:20
S. Deepak, et al. [7]	Deep CNN and Transfer Learning	98%	Axial, Coronal, and Sagittal	GoogLeNet	70:00:30
Proposed Solution	Few-shot transfer learning using a prototypical network	97.44%	Axial, Coronal, and Sagittal	ResNet34	70:20:10

Table 2: A summary table comprising of the researches [10, 14, 11, 7] and the proposed research comparing the models and elements including the architectures, accuracies, different angles used, pretrained networks and train/validation/test data split

test accuracy recorded significantly lower as a result. Therefore, dropout was applied. Dropout is not too appropriate in this instance as the dataset is so small and there are not many samples so it could have had an adverse effect, but it was necessary.

Finally, the model was limited by the computational resources provided by the free version of Google Colab. This proposal was created via Google Colab which limited the amount of RAM that could be used. Google Colab only allowed for the T4 processor for a short period every day or two and only in the T4 processor could features such as ResNet34 and slightly larger image sizes be used. Therefore, with the restrictions, testing and hyperparameter tuning were harder to achieve. But even with the T4 processor active, the small RAM sizes were not capable of handling ResNet50 or 512x512 images. To overcome this, the T4 GPU processor was used sparingly and only when testing was required, otherwise the standard CPU mode was active.

5.3 Future Work & Recommendations

Further testing and analysis could be applied in the future to experiment more and potentially make improvements on the chosen model. Other few-shot learning approaches could be tested such as meta-learning or Siamese networks to see if these make any changes to the proposed prototypical network. As well as this, the fine-tuning could be further tested and improved to better the performance of the model. In this proposed solution, only two linear layers were applied but for medical datasets, it is better to use deep fine-tuning where more layers are unfrozen and tuned instead of just the final fully classified layer as shallow transfer learning may be too generic and unable to capture the differences between medical images that deep transfer learning can [8]. This would further improve the performance as the model would be better suited to the target dataset. Also changing the pretrained model to a newer version, like ResNet50V2 as stated in the research paper by Md. Alamin Talukder, et al. [14], could also improve the performance of the model.

The model could not only just be improved through making changes to the model design or pretrained model but also the dataset. The dataset could be improved by getting more labelled examples which would improve the performance as more images can be trained on therefore improving the generalisability of the dataset and reducing the bias on one set if more images from the weaker classes were added. In addition, if the images could be labelled with the plane they are taken from (axial, sagittal, or coronal), these classes could be further distinguished. For example, there could be three prototypes for each class with each prototype being of a different plane. This would mean the embeddings are not too complicated and prototypes are not stretching far distances to fit all types of planes. This was utilised in the experiment by J. Paul, et al.[10] to only use one plane to make it simpler for the model and easier to learn.

Furthermore, the model could be tested on a larger variety of datasets including different kinds of brain tumours. This would help prove the generalisability of the model as well as show how it would work in real-world scenarios and if it could be used clinically for diagnosis in hospitals. The model could also be tested on a tougher dataset which contains images in which the model struggles on to show a true performance. These images would be the tougher, blurry and distorted images which could occur in a real hospital MRI scan and be costly to retake the scan.

6 Conclusion

This research set out to find a few-shot learning approach suitable for brain tumour classification from MRI scans. The chosen dataset consisted of three classes of brain tumour: meningioma, glioma and pituitary. To complete this project, a prototypical network was implemented with transfer learning to mitigate the minimal size of medical imaging datasets. Fine-tuning was applied to improve the performance of the feature extractor applied from the ResNet34 backbone. During training, episodic training was implemented with early stopping to quickly find the optimum number of epochs whilst also avoiding overfitting from running too many epochs. Many hyperparameters were altered and tuned to find the optimum values which result in the highest accuracy. These hyperparameters are number of shots, query size, number of episodes per epoch, gamma value of the focal loss function, the distance function, and other hyperparameters. In the end, the model was a 5-shot learning prototypical network using squared Euclidean distance algorithm with focal loss. This proposed solution resulted in a test accuracy of 97.44% which is comparable with other designs made in related work on the same dataset. This shows it is suitable for predicting brain tumour classifications under 3-way scenarios and could be useful for more classes as it is a few-shot learning model, so it is well adapted for learning new classes with only a few labelled samples. In the future, the dataset could be split into separate projection angles and tested on tougher images that the model is more likely to struggle on as to find its suitability for real-world instances where retaking an MRI scan image may be expensive and not viable due to time and cost restrictions. A few-shot learning model could be introduced as a potential solution to classifying brain tumours in hospitals. Brain tumours are sometimes hard to differentiate but artificial intelligence is well versed at identifying patterns and could be a better form of diagnosis than a well-trained doctor. Few-shot learning means only a few labelled staples need to be shown for the model to learn how to classify it. Also, through training on difficult images, the model can learn to identify tumours even if the MRI scan appears blurred or out of focus. This would save time and costs on retaking an MRI if a doctor cannot identify what is happening in the picture. This model was built using Google Colab which had many memory restrictions. To better improve the model, the backbone can be upgraded to a higher version of ResNet to allow for better feature extraction by training the model on a more powerful computer.

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